

ARTHUR JIUN WEI HWANG

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EDUCATION

BASc, Mechanical Engineering + PEY Co-op

Sep 2024 – May 2029 (Expected)

University of Toronto

- cGPA: 3.97 / 4.00

WORK EXPERIENCE

Library Development Member

Dec 2024 – Present

RocketPy Open Source, Remote

- Developed serialization functions to save and quickly retrieve Flight simulation data via a custom RPY file format.
- Implemented "create_environment_object" method for EnvironmentAnalysis class, enabling simulation Environment rendering from historical data across user-defined time intervals in a single function call.
- Improved parachute aerodynamic model accuracy by modeling it as a hemispheroid parameterized by radius, height, and porosity.

Summer Research Student

May 2025 – Aug 2025

Laboratory for Extreme Mechanics & Additive Manufacturing, Toronto, ON

- Developed [GLOW](#), an open-source GCODE generator for Direct Energy Deposition Additive Manufacturing, supporting 3 print geometries, 2 substrate configurations, 10 configurable print parameters, and 11 preprogrammed machine-specific codes.
- Deployed GLOW as a zero-dependency GUI desktop application using PyQt6 and PyInstaller, and distributed with Inno Setup, eliminating software dependency.
- Ported GLOW from Python to C++ using Qt Creator, reducing binary size from several GB to a lightweight executable and improving runtime performance by 40%, enabling practical deployment across lab workstations.
- Presented GLOW and GCODE generation methodology as guest speaker in the graduate-level MSE1068 Additive Manufacturing course at University of Toronto.

EXTRACURRICULAR ACTIVITIES

Aerodynamics Team Member

Sep 2024 – Present

University of Toronto Aerospace Team Rocketry Division, Toronto, ON

- Conducted aeroelasticity, fin flutter, and stability margin analyses in OpenRocket across dozens of nose cone and fin designs, achieving 1.5-3 caliber stability across 0-8 m/s wind conditions and 2x flutter safety factor.
- Built a RocketPy Monte Carlo simulation environment for 4 rocket projects, modeling manufacturing tolerances and environmental variability (including ballistic descent scenarios) to verify landing zones via KML export to Google Earth for safety compliance.
- Developed a Jupyter Notebook to automatically build Rocketpy simulations from OpenRocket files and scattered data sources (Excel, RSE files), reducing cross-validation tasks from several hours to minutes.
- Implemented a Genetic Algorithm Multidisciplinary Optimization framework integrated with RocketPy to automatically optimize aerosurface geometry, replacing manual parameter sweeps.

Aerodynamics Team Member

Apr 2024 – Aug 2024

Keep Flying Poli USP Jr, Sao Paulo, SP

- Designed a micro class aircraft with 2.16kg MTOW able to take off in 4 m runway for SAE Brasil Aerodesign 2024.
- Developed Julia programs for custom airfoil parametrization and generation, Sobol global sensitivity analysis of 10 design variables on MTOW, critical section analysis, and aircraft performance robustness test.
- Co-authored two sections of the SAE Brasil Aerodesign 2024 aerodynamics report and led a word reduction pass cutting word count by 5% to meet submission requirements.

SKILLS

Programming Languages: Python (Advanced), [MATLAB](#) (Advanced), C++ (Intermediate), Julia (Intermediate)

Technical Tools: Microsoft Office Suite (Advanced), SolidWorks (Advanced), VS Code (Advanced), GitHub (Advanced), Siemens NX (Intermediate), [Simulink](#) (Intermediate)

Languages: Portuguese (Native), English (Advanced), Chinese (Intermediate)